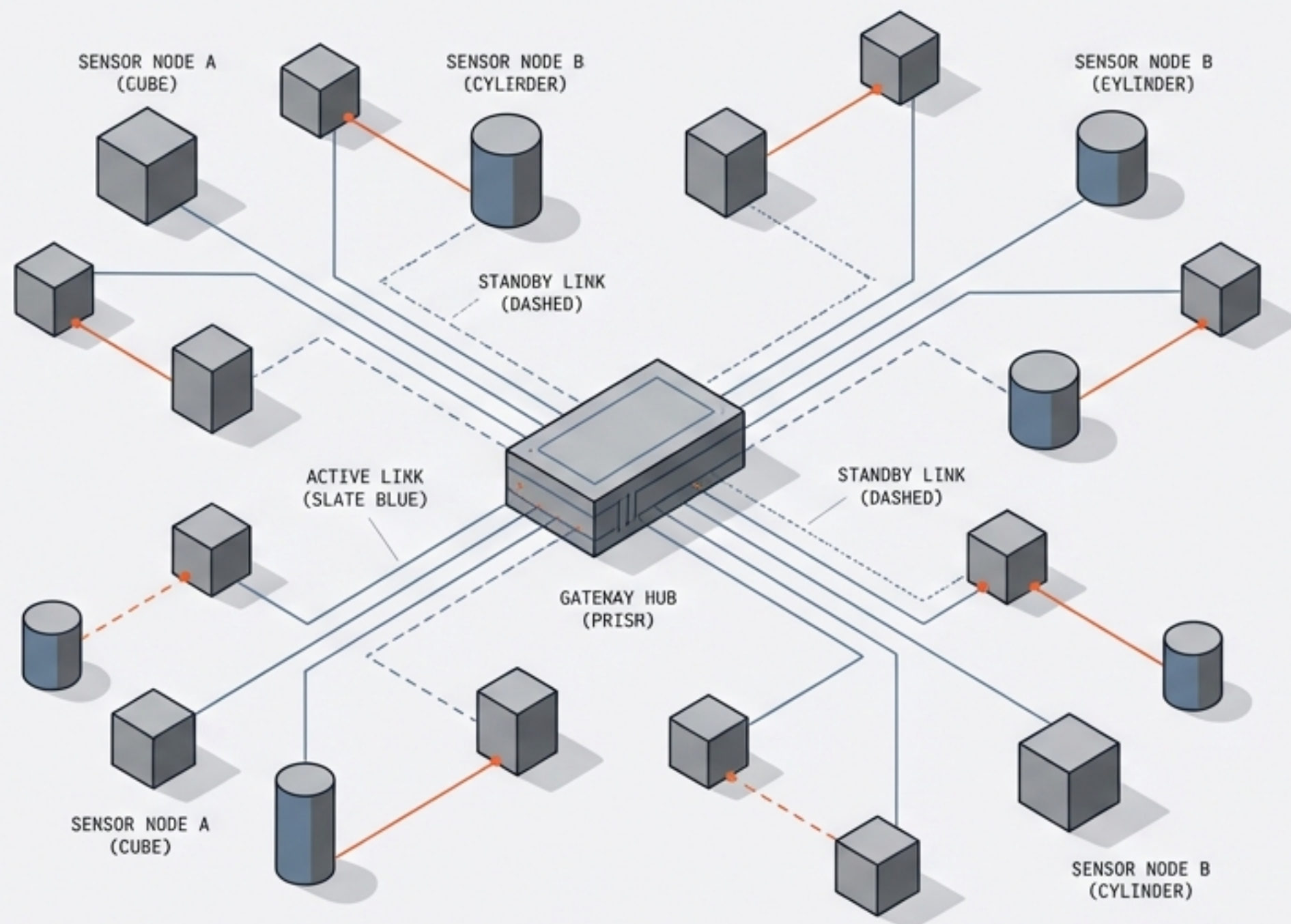




HetOTA: Secure Lifecycle Management for Heterogeneous IoT Networks

A lightweight, vendor-agnostic framework bridging **ESP32** and **Raspberry Pi** environments with **Post-Quantum Cryptography**.

Bridging the Security Gap in Mixed-Architecture Networks

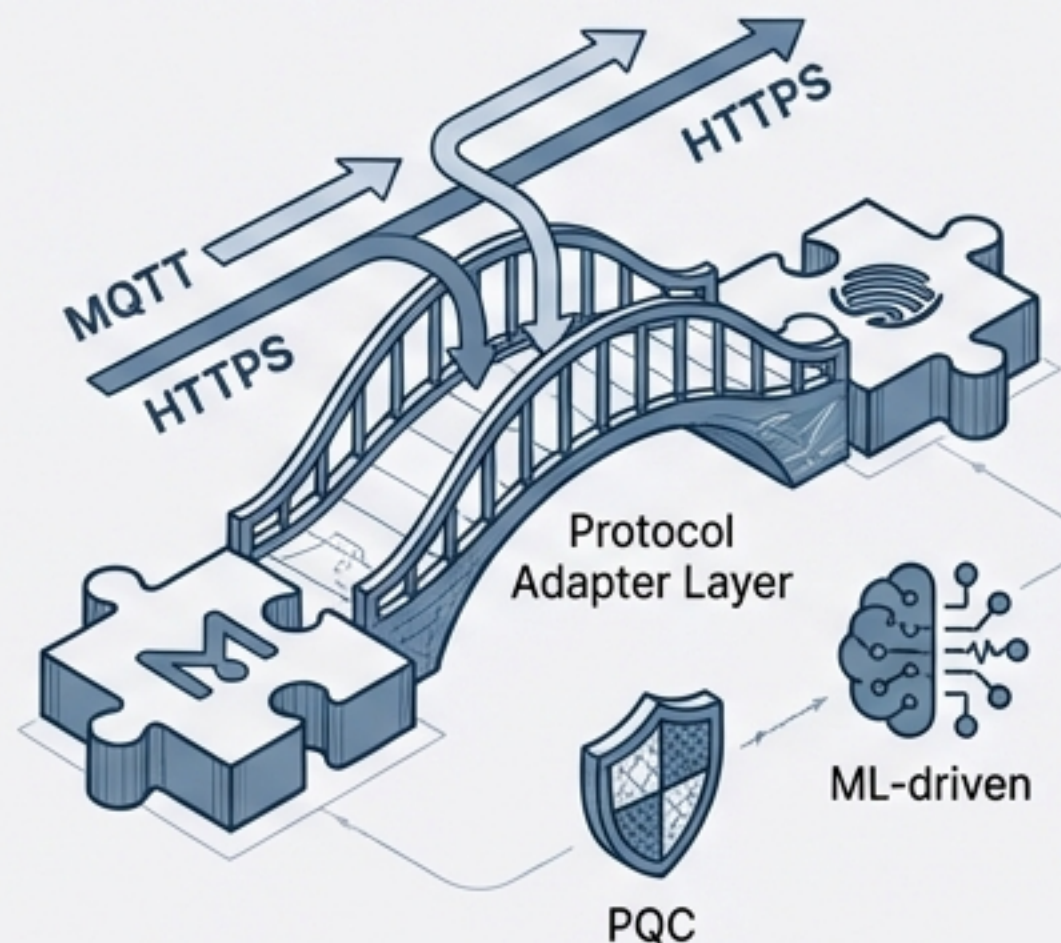
The Gap

Current OTA solutions are siloed. Tools like Mender cover gateways, while ESP-IDF covers sensors. There is no unified, scalable framework for "brownfield" (legacy/mixed) networks involving both resource-constrained microcontrollers and Single Board Computers (SBCs).



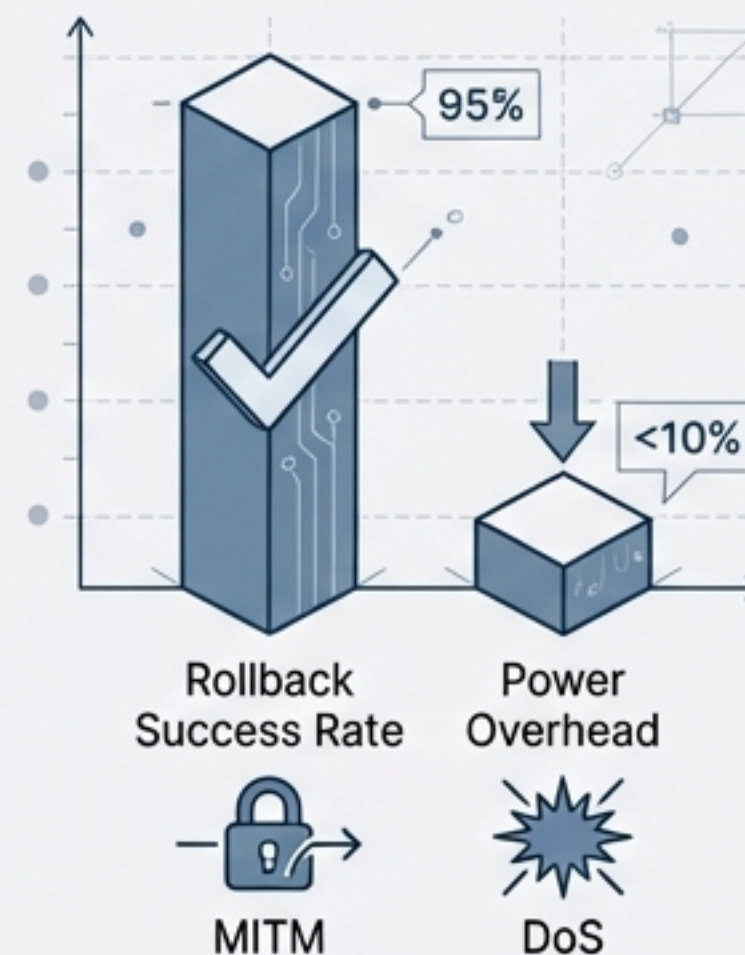
The Solution (HetOTA)

A novel framework introducing a Protocol Adapter Layer to abstract communication (MQTT/HTTPS). It integrates lightweight Post-Quantum Cryptography (PQC) and ML-driven anomaly detection.



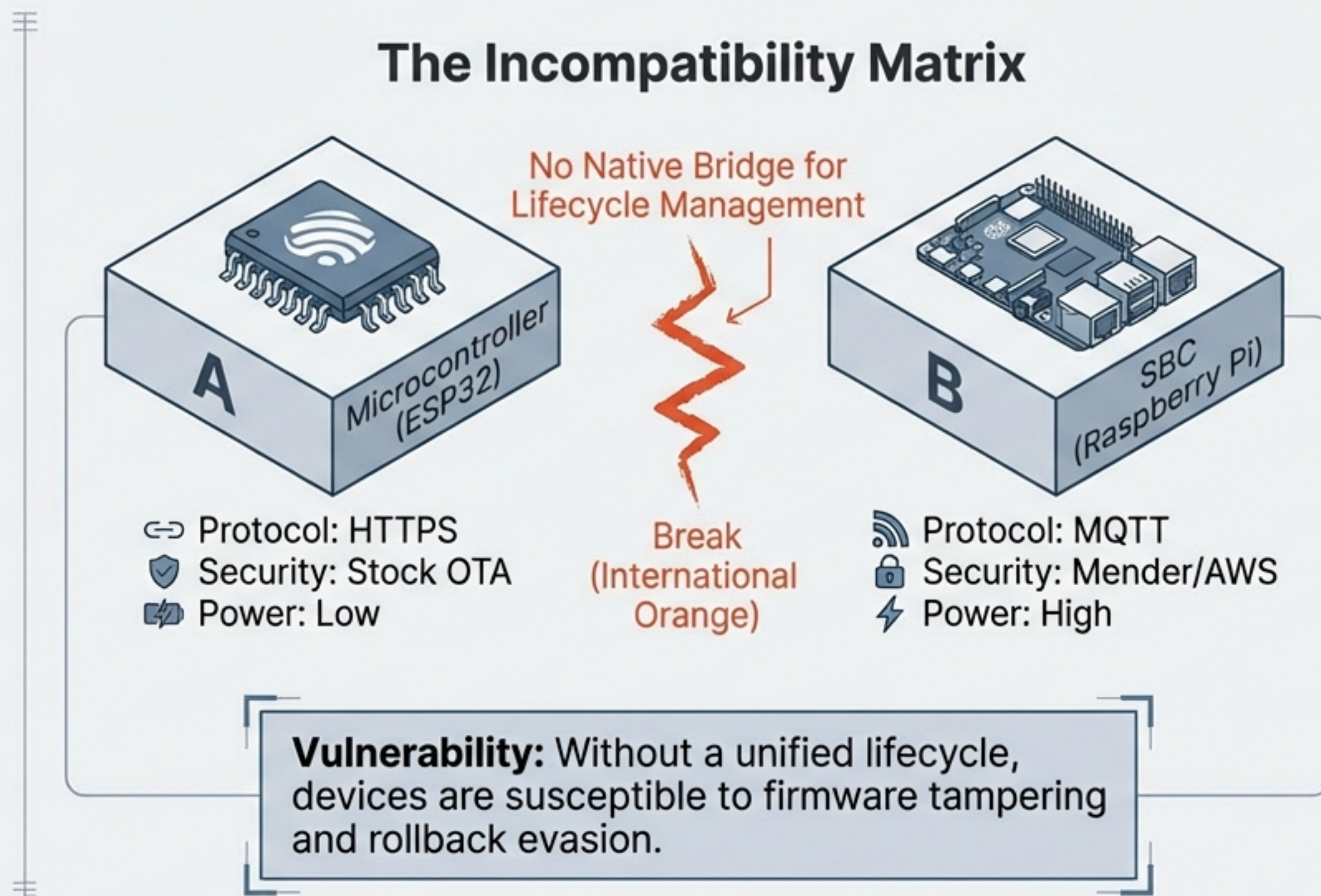
The Validation

Validated via attack simulations (MITM, DoS) achieving a 95% rollback success rate with <10% power overhead on ESP32 endpoints.

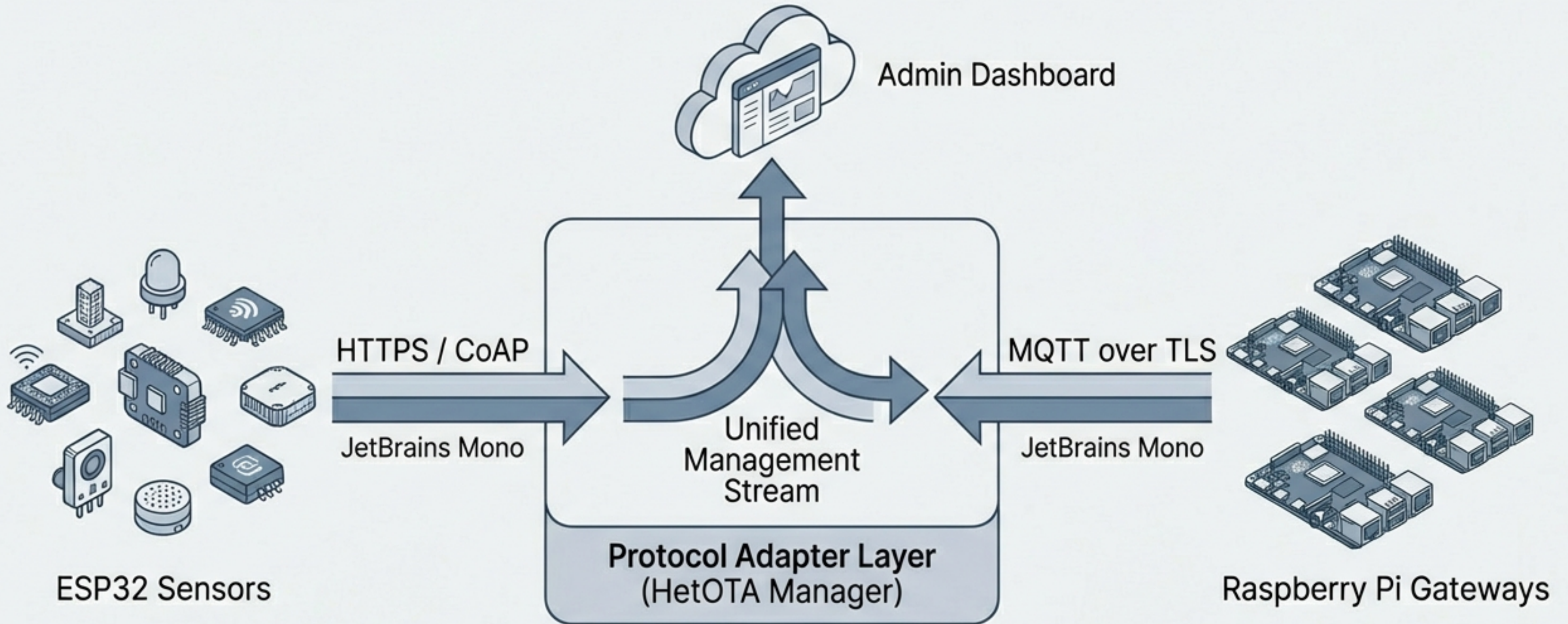


The Risks of Siloed Security in Brownfield Deployments

The IoT ecosystem suffers from a “deploy-and-forget” mentality. While new devices are secure, secure, existing mixed fleets lack consistent management.



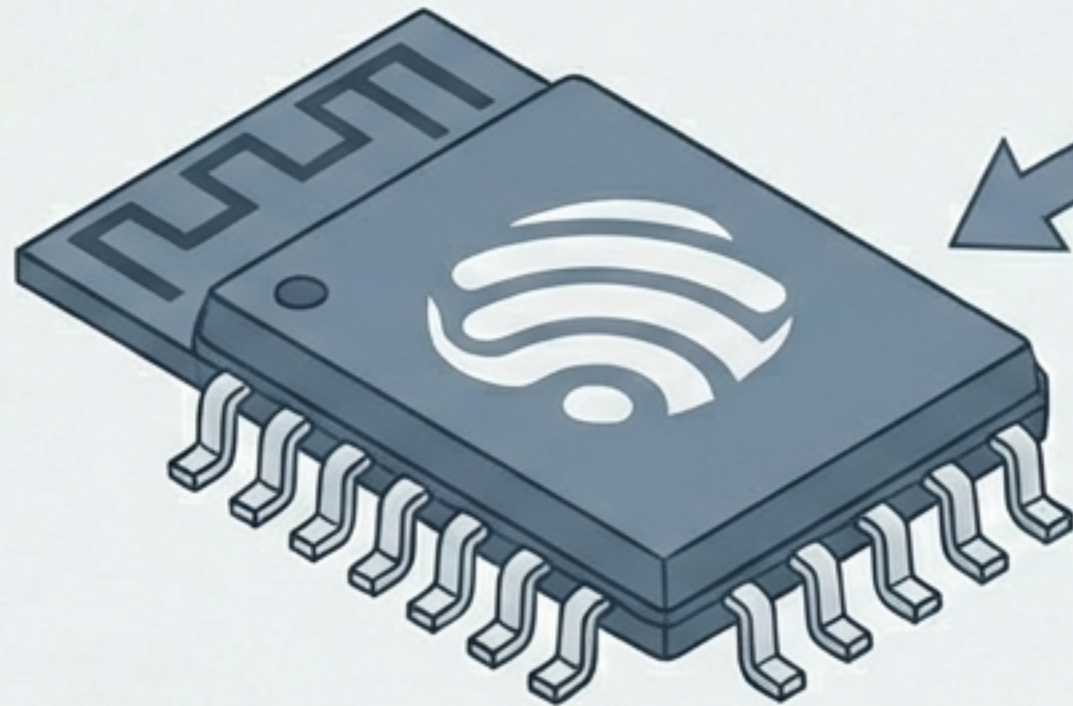
The HetOTA Framework: A Protocol-Agnostic Architecture



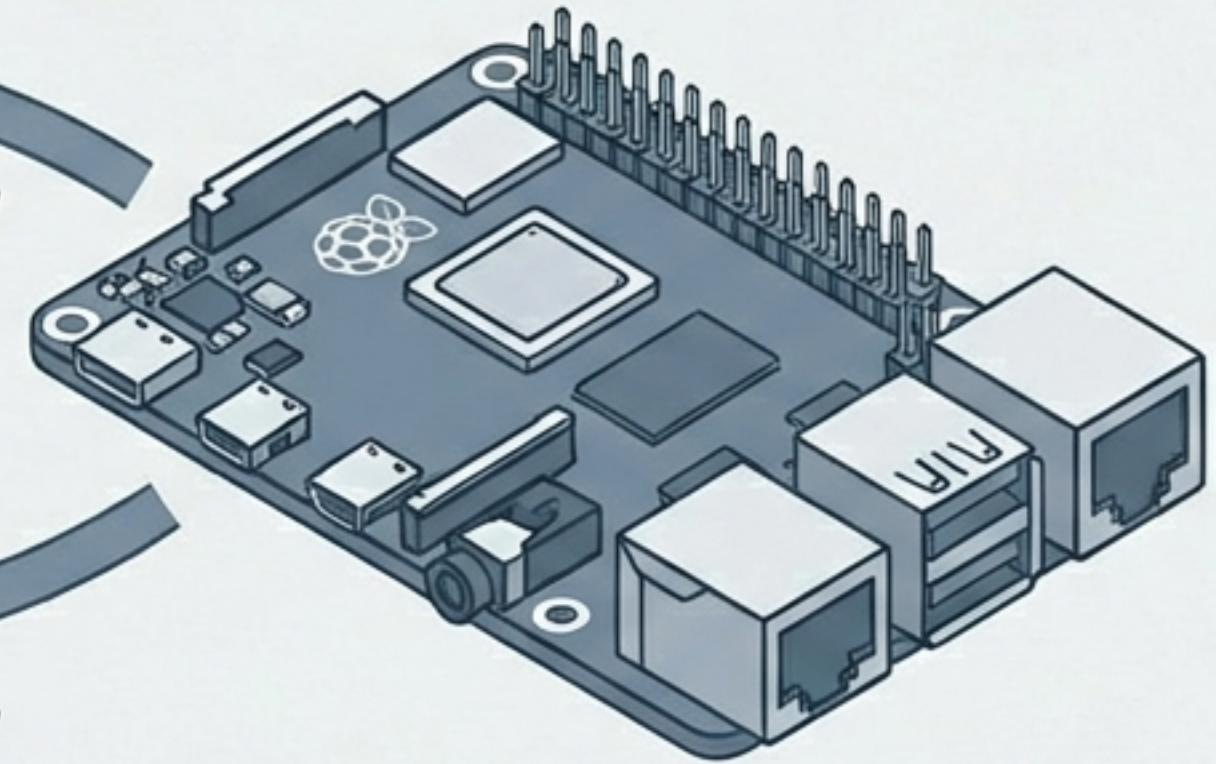
By abstracting protocols into a central manager, HetOTA solves the core issue of heterogeneity.

Hardware Roles: Balancing Constraint with Intelligence

ESP32 (The Sensor Node)



Raspberry Pi (The Gateway/Server)



Management Flow

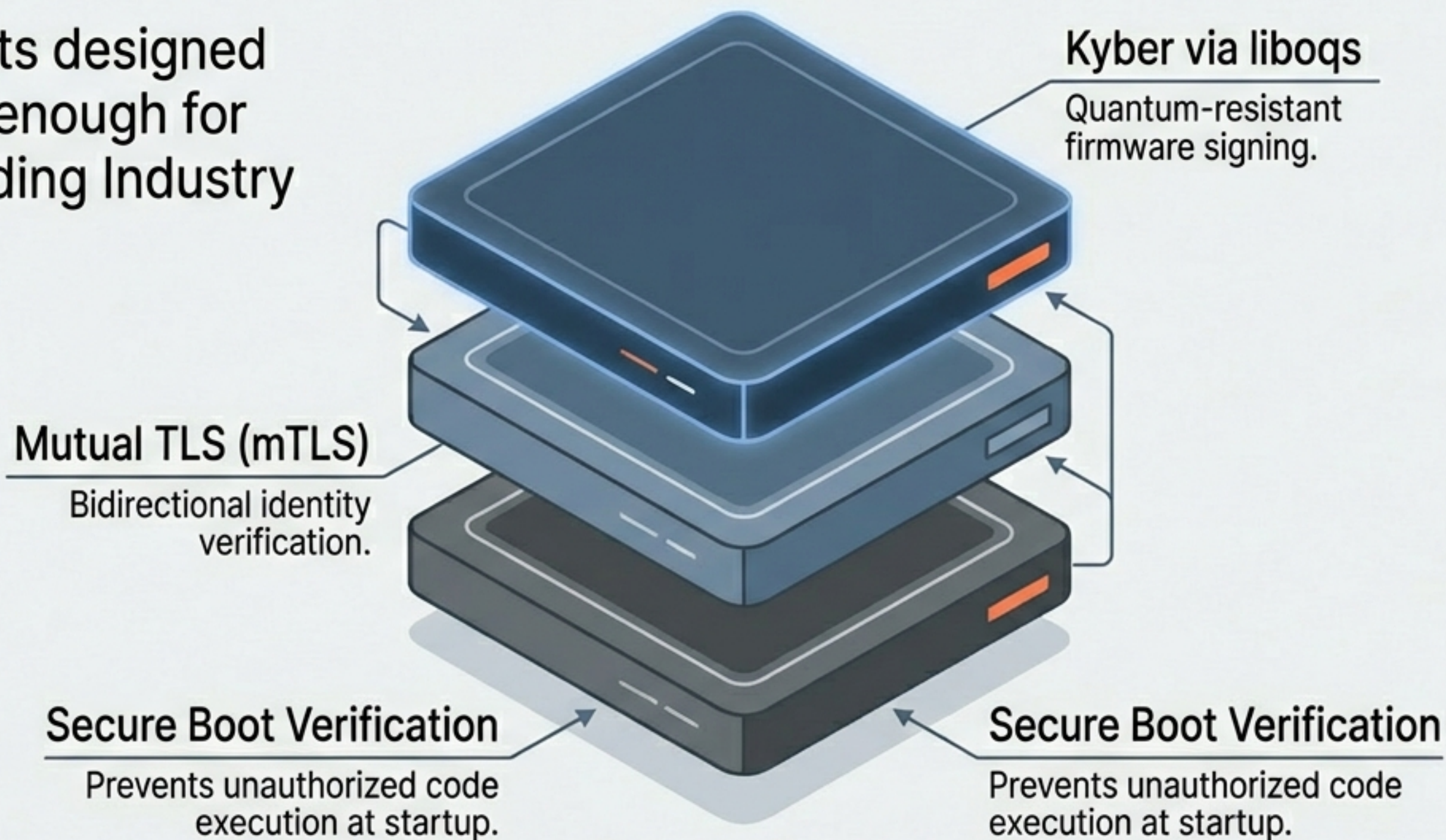
Management Flow

- Role: OTA Client
- Storage: NVS-secure certificate storage
- Task: Runs TinyML for local power baselines

- Role: Dockerised Manager & MQTT Broker
- Task: Python-based payload analysis and anomaly detection

Future-Proofing with Lightweight Post-Quantum Cryptography

Core Stack Components designed to remain **lightweight** enough for the ESP32 while providing Industry 5.0 grade security.



Phase 1: Trust Establishment and Onboarding

Provisioning



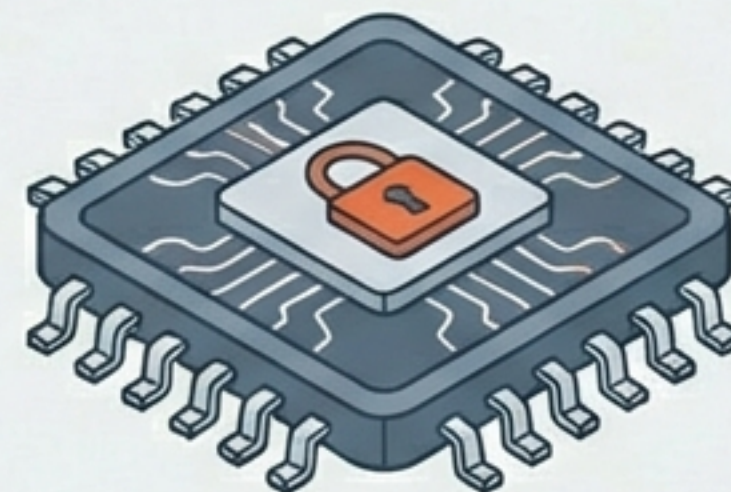
Initial connection established via BLE.

Identity



QR-code based certificate issuance.

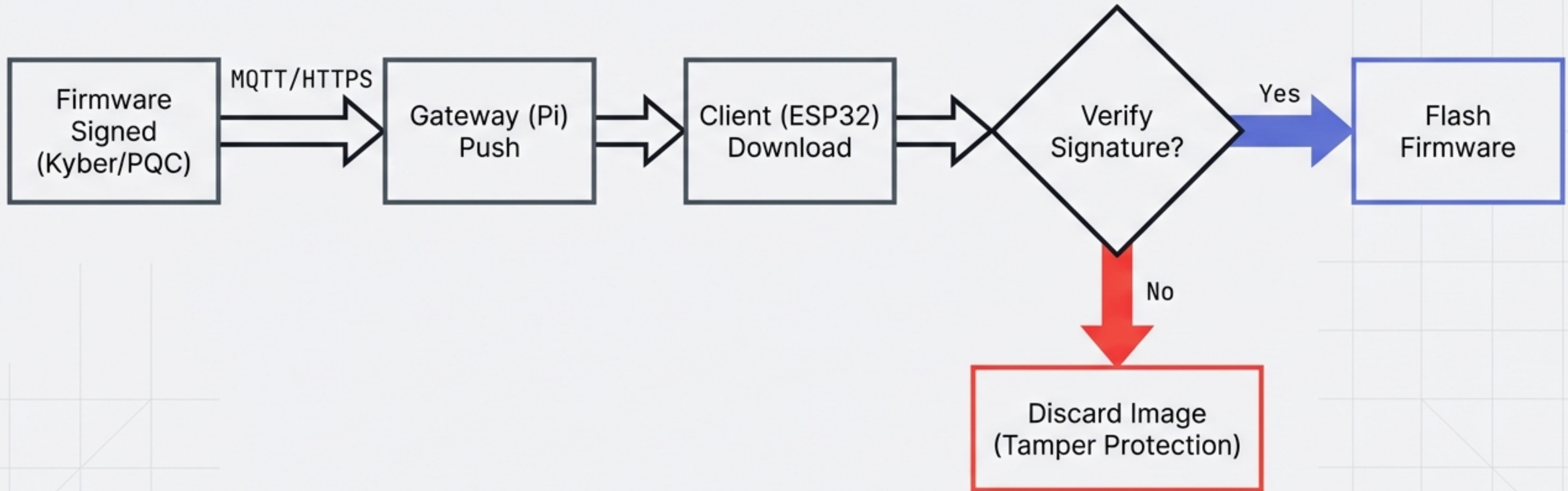
Storage



Certificates stored in ESP32 NVS (Non-Volatile Storage).

Outcome: Root of Trust established without hardcoding secrets.

Phase 2: Secure OTA Delivery and Verification



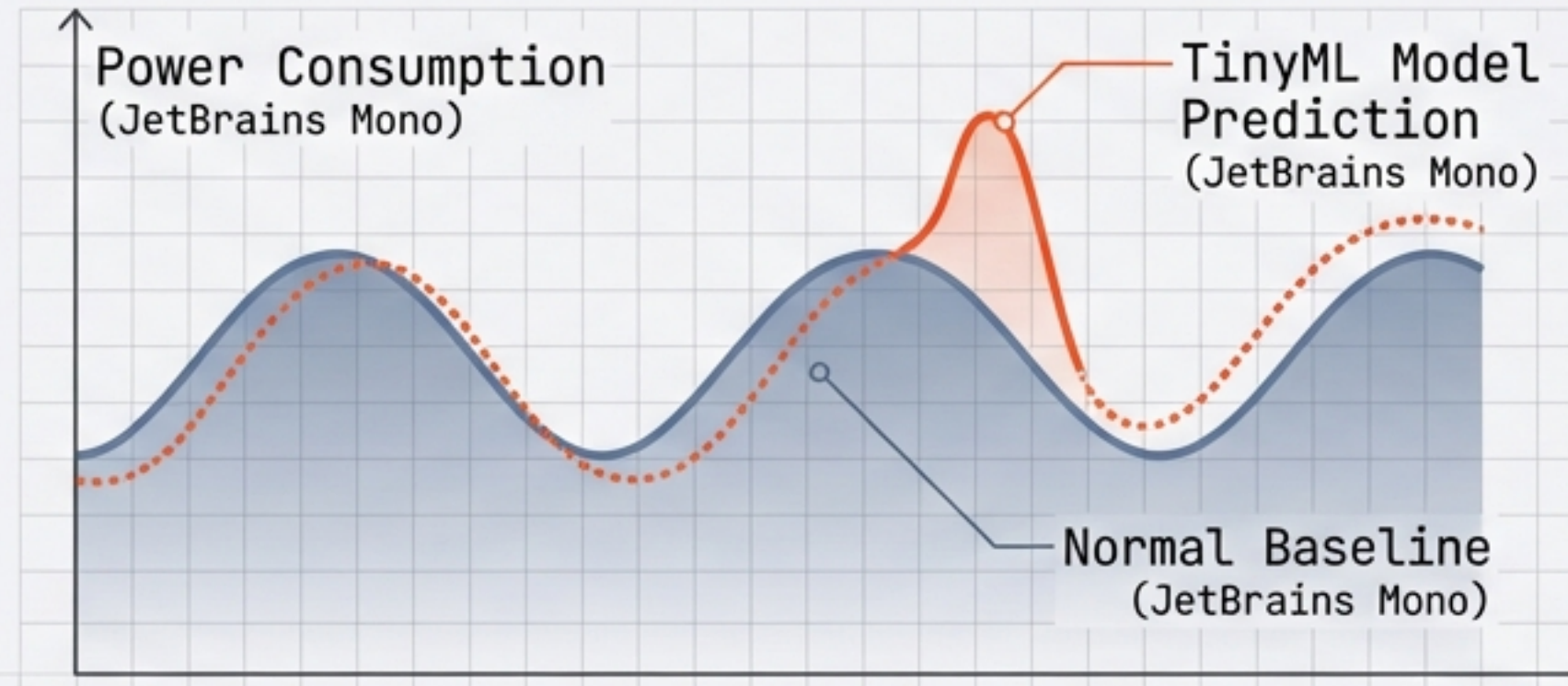
Phase 3: Continuous Monitoring via Edge ML

Gateway Analysis (Raspberry Pi)

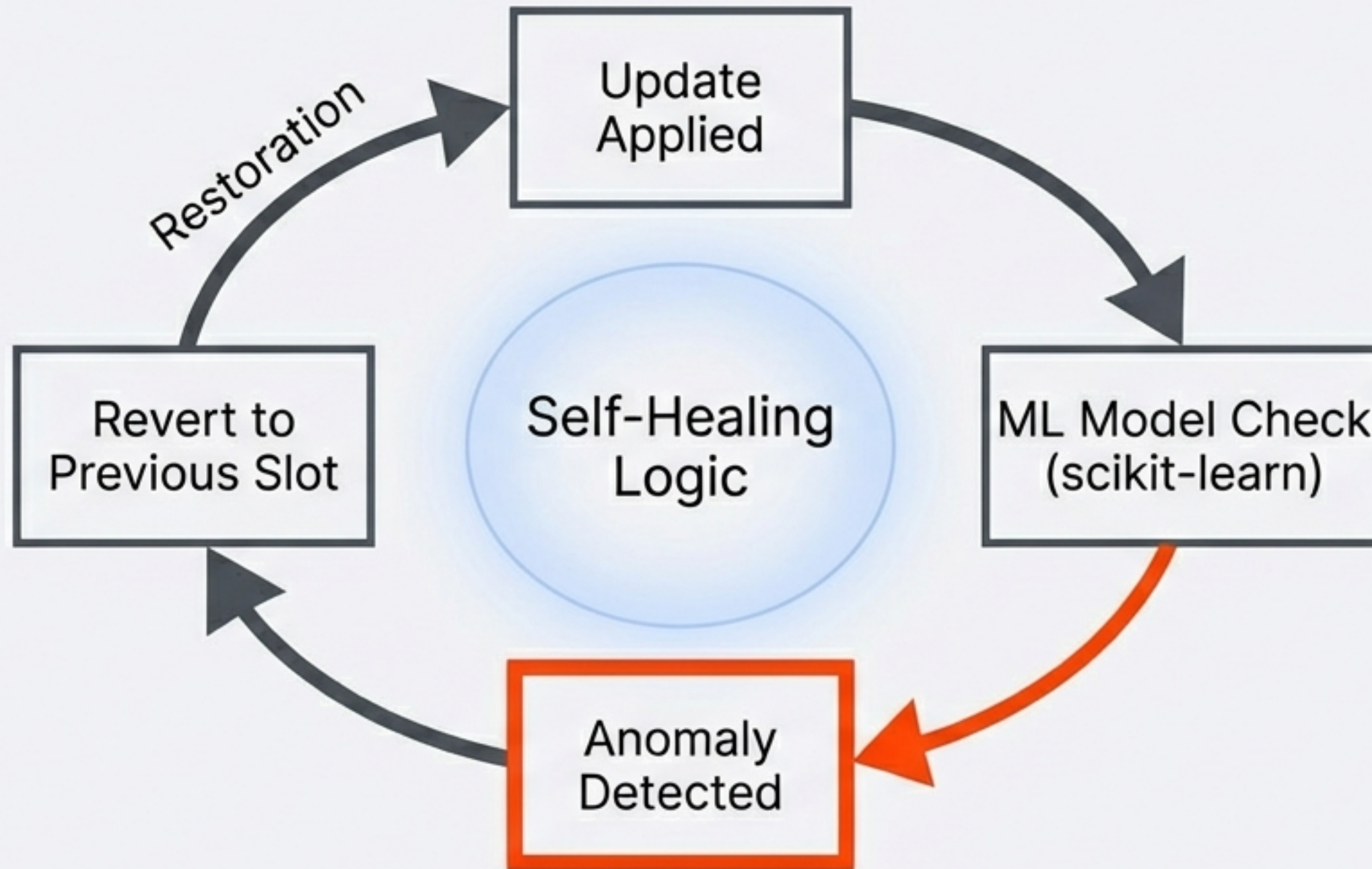


Detecting “soft failures”—updates that install successfully but cause behavioural issues like battery drain.

Edge Baselines (ESP32)



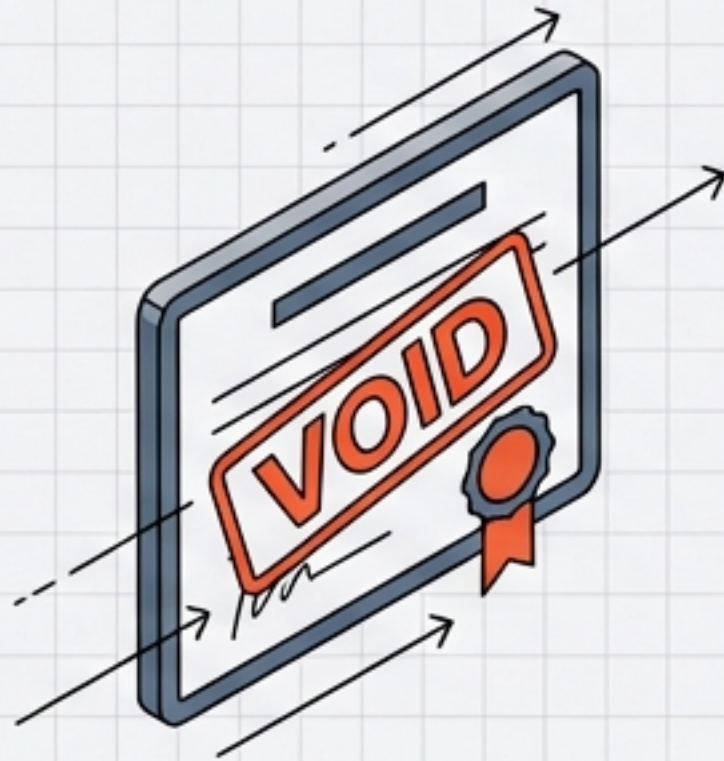
The Safety Net: Automated Rollback Logic



Ensures a bad update never bricks a remote device.

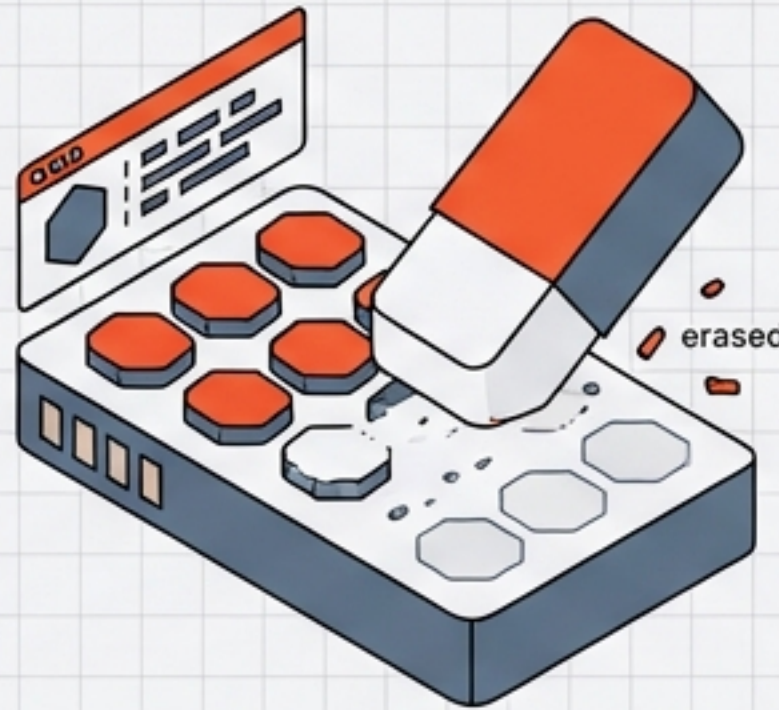
Phase 4: Secure Decommissioning and Zeroisation

When a device reaches End-of-Life (EOL), simple physical removal is insufficient due to stored credentials.



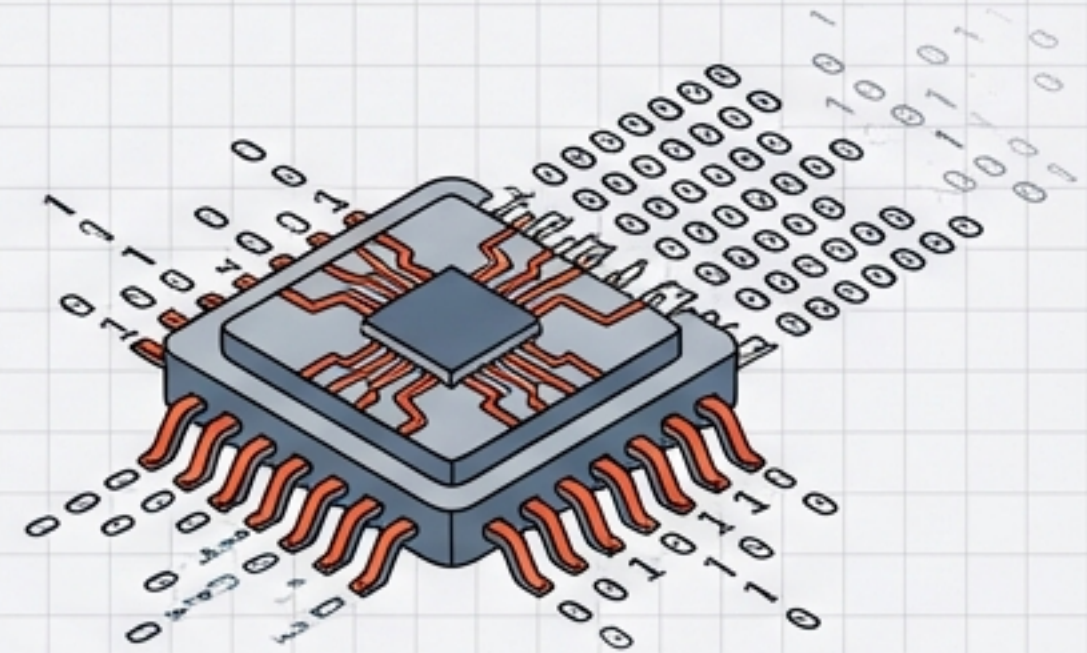
Revocation

Certificates formally revoked at Gateway.



Wiping

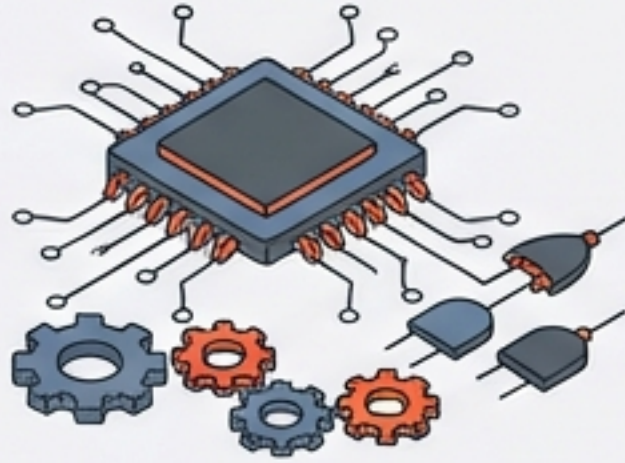
Keys wiped from NVS.



Zeroisation

Full device zeroisation to prevent physical recovery.

The Implementation Stack



Firmware & Logic

C (ESP32) /
Python (Gateway)



Communication

Mosquitto (MQTT
Broker)

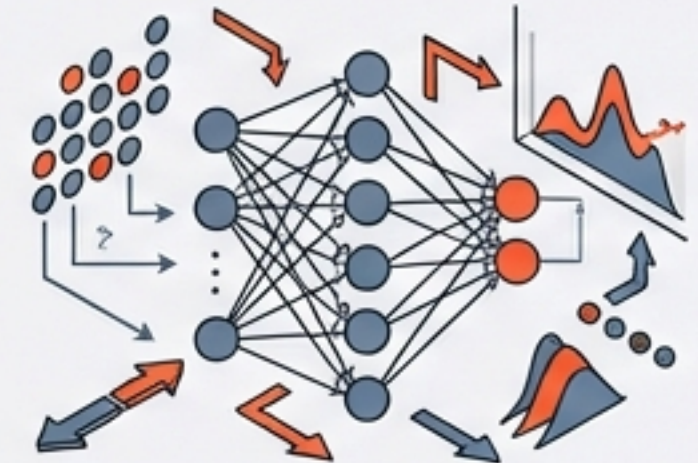


Containerisation

Docker
(Gateway Services)

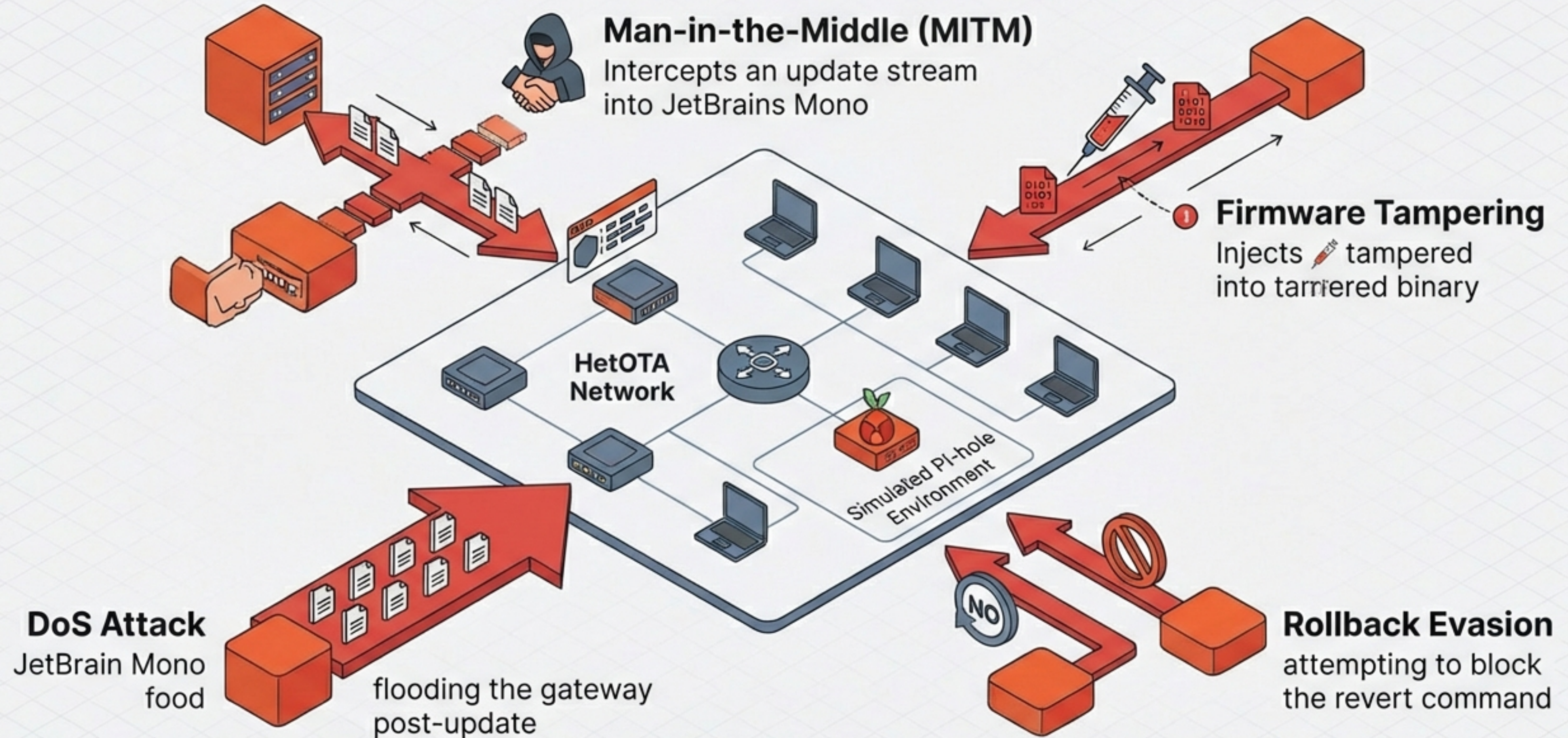
Machine Learning

Scikit-learn
(Anomaly Detection)



Environment: Developed and tested within a Parrot OS / Raspberry Pi setup.

Adversarial Simulation and Stress Testing

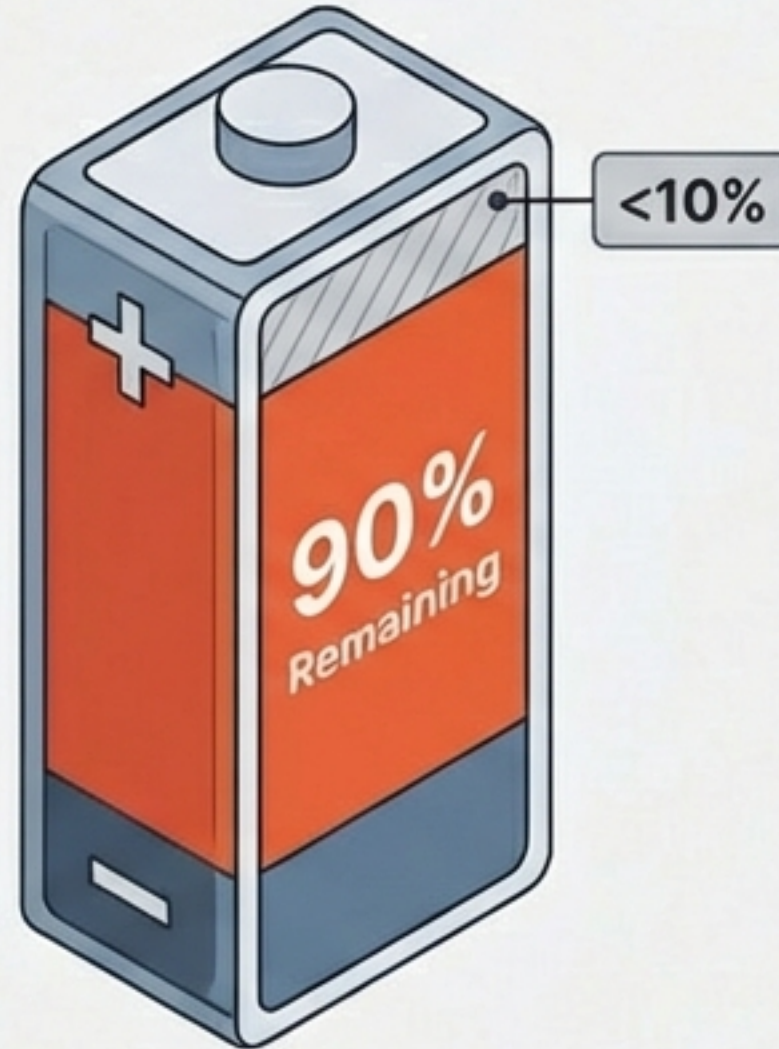


Evaluation Metrics: Efficiency vs Security



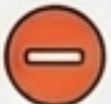
**95% Rollback
Success Rate**

In simulated attack scenarios.



**<10% Power
Overhead**

On ESP32 endpoints.

Solution	Performance	Mixed-Fleet Capable
HetOTA	High	 Yes
AWS IoT	Medium	 Limited
Standard ESP OTA	Low	 No

From Prototype to Industry 5.0 Standard

HetOTA provides the first practical prototype for mixed MCU/SBC fleets that operates under real-world resource constraints, extending existing surveys by Sharma et al.

Secure. Scalable. Heterogeneous.

